

Evaluating Susceptibility to Pancreatic Cancer: ASCO Provisional Clinical Opinion

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PURPOSE An ASCO provisional clinical opinion (PCO) offers timely clinical direction to ASCO's membership and other health care providers. This PCO addresses identification and management of patients and family members with possible predisposition to pancreatic adenocarcinoma.

METHODS ASCO convened an Expert Panel and conducted a systematic review of the literature published from January 1998 to June 2018. Results of the databases searched were supplemented with hand searching of the bibliographies of systematic reviews and selected seminal articles and contributions from Expert Panel members' curated files.

PROVISIONAL CLINICAL OPINION All patients diagnosed with pancreatic adenocarcinoma should undergo assessment of risk for hereditary syndromes known to be associated with an increased risk for pancreatic adenocarcinoma. Assessment of risk should include a comprehensive review of family history of cancer. Individuals with a family history of pancreatic cancer affecting two first-degree relatives meet criteria for familial pancreatic cancer (FPC). Individuals (cancer affected or unaffected) with a family history of pancreatic cancer meeting criteria for FPC, those with three or more diagnoses of pancreatic cancer in same side of the family, and individuals meeting criteria for other genetic syndromes associated with increased risk for pancreatic cancer have an increased risk for pancreatic cancer and are candidates for genetic testing. Germline genetic testing for cancer susceptibility may be discussed with individuals diagnosed with pancreatic cancer, even if family history is unremarkable. Benefits and limitations of pancreatic cancer screening should be discussed with individuals whose family history meets criteria for FPC and/or genetic susceptibility to pancreatic cancer.

Additional information is available at www.asco.org/gastrointestinal-cancer-guidelines.

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INTRODUCTION

ASCO has established an approach to offer a rapid response to emerging information in clinical oncology. The provisional clinical opinion (PCO) is intended to offer timely clinical direction to ASCO's oncologists and other health care providers, including primary care physicians. This PCO was prioritized to address a research area where there are emerging data but not a robust landscape of clinical trials. The Expert Panel includes a discussion section on limitations of the research to guide future directions. This PCO should be read with the understanding that randomized clinical trial data are not available for these guidance statements, but it is the opinion of the Expert Panel that the statements made represent the state of the data available.

This PCO addresses identification and management of patients and family members with predisposition to pancreatic adenocarcinoma. Estimates suggest up to 10% of all pancreatic adenocarcinoma cases are

familial,¹ and pathogenic germline variants in specific genes have been associated with lifetime risks of pancreatic cancer ranging from 4% to 40%.²⁻⁵ For these high-risk individuals, surveillance offers the potential for early identification of pancreatic neoplasms.^{1,2} This PCO addresses how susceptibility to adenocarcinomas of the pancreas should be assessed, who should be genetically tested and/or screened for familial predisposition to pancreatic adenocarcinoma, and what pancreas surveillance strategies should be used in individuals with predisposition to pancreatic ductal adenocarcinoma.

METHODS

This systematic review-based product was developed by a multidisciplinary Expert Panel, which included a patient representative and ASCO guidelines staff with health research methodology expertise. Computerized literature searches of PubMed and the Cochrane Collaboration Library were performed. The searches of

ASSOCIATED CONTENT

Appendix

Data Supplement

Author affiliations and support information (if applicable) appear at the end of this article.

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Clinical Practice Guidelines

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Editor's note: Additional information, including a Data Supplement with additional evidence tables, a Methodology Supplement, slide sets, clinical tools and resources, and links to patient information at www.cancer.net, is available at www.asco.org/gastrointestinal-cancer-guidelines.

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THE BOTTOM LINE

Evaluating Susceptibility to Pancreatic Cancer: ASCO Provisional Clinical Opinion

Research Question

How can individuals at increased risk for pancreatic cancer be identified and managed?

Target Population

People diagnosed with pancreatic adenocarcinoma and families or individuals with concern for genetic predisposition to pancreatic cancer.

Target Audience

Primary care physicians, medical oncologists, nurse practitioners, surgeons, gastroenterologists, internists, and other health care providers.

Methods

ASCO convened an Expert Panel and conducted a systematic review of the literature published from January 1998 to June 2018. Results of the database searches were supplemented with hand searching of the bibliographies of systematic reviews and selected seminal articles and contributions from Expert Panel members' curated files.

Provisional Clinical Opinion

Research Question 1

How should susceptibility for pancreatic cancer be assessed? What is the role of family history of cancer? Which individuals are considered as having predisposition to pancreatic cancer?

PCO 1.1 Clinical evaluations of people (with or without pancreatic cancer) should include assessment for possible genetic predisposition syndromes, beginning with a review of family history of cancer (including tumor types and ages at diagnosis for all first- and second-degree relatives)⁴³ (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

PCO 1.2 Individuals with a family history of pancreatic cancer affecting two first-degree relatives meet criteria for familial pancreatic cancer. Individuals whose family history meets criteria for familial pancreatic cancer, those with three or more diagnoses of pancreatic cancer in same side of the family, and individuals meeting criteria for other genetic syndromes (Table 1) associated with increased risk for pancreatic cancer have an increased risk for pancreatic cancer and are candidates for genetic testing (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

Qualifying Statement. It is important to note that for 90% of families meeting criteria for familial pancreatic cancer, genetic testing does not detect a pathogenic mutation; therefore, there may be additional shared epigenetic, genetic, or environmental factors that contribute to pancreatic cancer risk.

PCO 1.3 Genetic risk evaluation should be conducted in conjunction with health care providers familiar with the diagnosis and management of hereditary cancer syndromes to determine the most appropriate testing strategy and discuss implications of the findings for family members. Germline genetic testing for patients with pancreatic cancer should be offered in the context of shared decision making⁴⁴⁻⁴⁷ (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

Research Question 2

Which individuals should undergo genetic testing for predisposition to pancreatic cancer?

PCO 2.1 All patients diagnosed with pancreatic adenocarcinoma should undergo assessment of risk for hereditary syndromes known to be associated with an increased risk for pancreatic adenocarcinoma (Table 1). Assessment of risk includes obtaining a personal cancer history and family history of cancers in first- and second-degree relatives. However, recent data demonstrate that many individuals who develop pancreatic cancer in the setting of genetic predisposition lack clinical features or family cancer history typically associated with the corresponding hereditary syndrome. Therefore, germline genetic testing may be discussed with patients with personal history of pancreatic cancer, even if family history is unremarkable (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

PCO 2.2 An individual with a cancer diagnosis is often the best candidate in whom to initiate genetic testing and has the highest likelihood of an informative test result; however, if a cancer-affected individual is not available, testing may be performed in a pancreatic cancer-unaffected individual following genetic risk assessment, with the understanding that a negative test result is considered clinically uninformative.

The following cancer-unaffected individuals should be offered genetic risk evaluation:

- Members of families with an identified pathogenic cancer susceptibility gene variant
(continued on following page)

THE BOTTOM LINE (CONTINUED)

- Pancreatic cancer–unaffected individuals from families that meet criteria for genetic evaluation for known hereditary syndromes that are linked to pancreatic cancer
- Pancreatic cancer–unaffected individuals from families that meet criteria for familial pancreatic cancer, as outlined in PCO 1.2

(Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

PCO 2.3 Genetic testing in a patient with pancreatic cancer may confirm the diagnosis of a hereditary cancer syndrome and inform management of at-risk family members. Given the possibility that certain germline variants could potentially be used to guide therapeutic decision making and the limited prognosis of many patients with pancreatic cancer, the Expert Panel recommends that consideration of germline testing for inherited cancer susceptibility should be performed early in the disease course for patients with pancreatic cancer (Type: informal consensus; relative balance of benefits and harms; Strength of statement: moderate).

PCO 2.4 Several genes have been linked to risk for pancreatic cancer (Table 1). Unless a genetic diagnosis has previously been confirmed in a family member, germline genetic testing should be performed using a multigene panel that includes the genes listed in Table 1. A finding of a pathogenic or likely pathogenic germline variant can confer increased risks of cancers beyond the pancreas for the probands and their families; finding a germline variant of uncertain significance is not considered to be causative of increased cancer susceptibility^{47,51} (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

Research Question 3

What surveillance strategies should be used for individuals with predisposition to pancreatic ductal adenocarcinoma to screen for pancreatic and other cancers?

PCO 3.1 Pancreatic cancer surveillance can be considered for individuals who are first-degree relatives of individuals with familial pancreatic cancer^{2,5,34,37,52} and/or individuals with a family history of pancreatic cancer who carry a pathogenic germline variant in genes associated with predisposition to pancreatic cancer (Table 1). The potential risks, benefits, uncertainties, and limitations of surveillance for pancreatic cancer should be discussed in detail with individuals who are being considered for pancreatic cancer surveillance prior to beginning such surveillance. When possible, pancreatic surveillance should be performed at centers with the appropriate expertise to manage individuals at increased risk for pancreatic cancer. Surveillance may be performed with various modalities, including pancreas protocol magnetic resonance imaging/magnetic resonance cholangiopancreatography and/or endoscopic ultrasound. There are currently no approved biomarkers for screening and surveillance. CA 19-9 is not recommended as a screening test in the general population due to low specificity and sensitivity; its potential utility in pancreatic screening of high-risk individuals has not been established (Type: informal consensus; relative balance of benefits and harms; Strength of statement: moderate).

Qualifying Statement. Although large studies confirming mortality benefit of pancreatic screening are lacking, emerging data suggest screening in individuals at high risk is associated with downstaging of incident cancers.⁵³

PCO 3.2 There is not yet consensus on pathologic targets for surveillance, but the Expert Panel agrees the ultimate goal should be detection and treatment of high-grade dysplasia to prevent invasive cancer (Type: informal consensus; relative balance of benefits and harms; Strength of statement: moderate).

PCO 3.3 The potential risks of surveillance, including the risk of overtreatment and unnecessary resections, should be discussed with the patient. Given the challenges, patients should optimally be managed by an expert multidisciplinary team with experience in pancreatic cancer surveillance. Additional clinical studies are needed to determine the optimal approach for pancreatic surveillance (Type: informal consensus; relative balance of benefits and harms; Strength of statement: moderate).

PCO 3.4 There is currently a lack of consensus regarding which lesions discovered by pancreatic imaging require resection. Findings that generally warrant resection include lesions that are solid and/or that are associated with obstructive jaundice and/or dilation of the main pancreatic duct greater than 10 mm. The presence of worrisome features (cyst size greater than 3 cm, thickened/enhancing walls, mural nodule, dilated main pancreatic duct greater than 5 mm, abrupt change in duct caliber, or rapid growth) and the presence of three or more pancreatic cysts in the pancreas of high-risk individuals are associated with an increased risk of neoplastic progression.⁵³

Additional Resources More information, including a Data Supplement, a Methodology Supplement with information about evidence quality and strength of PCOs, slide sets, and clinical tools and resources, is available at www.asco.org/gastrointestinal-cancer-guidelines. Patient information is available at www.cancer.net

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

the English-language literature published from January 1998 to January 2018 combined hereditary pancreatic neoplasm terms and screening and intervention-related terms and MeSH headings. Results of the database searches were supplemented with hand searching of the bibliographies of systematic reviews and selected seminal articles and contributions from Expert Panel members' curated files through June 2018. The Expert Panel met via teleconference and e-mail to formulate the research questions and the recommendations. The guideline recommendations were sent for an open comment period of 2 weeks, allowing the public to review and comment on the recommendations after submitting a confidentiality agreement. These comments were taken into consideration while finalizing the recommendations. The draft manuscript and supplements were circulated to the panel for review and approval. The final manuscript was reviewed and approved by the ASCO Clinical Practice Guidelines Committee.

The members of the PCO Expert Panel are listed in Appendix Table A1 (online only). ASCO PCOs are updated by the Expert Panel on the basis of periodic review and analysis of new, potentially practice-changing information on the topic. All funding for the administration of the project was provided by ASCO.

Guideline Disclaimer

The clinical practice guidelines and other guidance published herein are provided by the American Society of Clinical Oncology, Inc. (ASCO) to assist providers in clinical decision making. The information therein should not be relied upon as being complete or accurate, nor should it be considered as inclusive of all proper treatments or methods of care or as a statement of the standard of care. With the rapid development of scientific knowledge, new evidence may emerge between the time information is developed and when it is published or read. The information is not continually updated and may not reflect the most recent evidence. The information addresses only the topics specifically identified therein and is not applicable to other interventions, diseases, or stages of diseases. This information does not mandate any particular course of medical care. Furthermore, the information is not intended to substitute for the independent professional judgment of the treating provider, as the information does not account for individual variation among patients. Recommendations reflect high, moderate or low confidence that the recommendation reflects the net effect of a given course of action. The use of words like "must," "must not," "should," and "should not" indicate that a course of action is recommended or not recommended for either most or many patients, but there is latitude for the treating physician to select other courses of action in individual cases. In all cases, the selected course of action should be considered by the treating provider in the context of treating the

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PCO and Conflict of Interest

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines ("Policy," found at <http://www.asco.org/rwc>). All members of the Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Panel did not disclose any relationships constituting a conflict under the Policy.

RESULTS

A total of 40 papers met eligibility criteria and form the evidentiary basis for the PCO recommendations.^{2,3,5-42} The papers informed the panel members but, ultimately, did not establish a strong evidence base to craft the recommendations. The results of the literature search identified the gaps in knowledge on the topic, which led to the section highlighting the limitations of the research and future directions. Additional information can be found in the Data Supplement.

ASCO'S PROVISIONAL CLINICAL OPINION

Research Question 1

How should susceptibility for pancreatic cancer be assessed? What is the role of family history of cancer? Which individuals are considered as having predisposition to pancreatic cancer?

PCO 1.1 Clinical evaluations of people (with or without pancreatic cancer) should include assessment for possible genetic predisposition syndromes, beginning with a review of family history of cancer (including tumor types and ages at diagnosis for all first- and second-degree relatives)⁴³ (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

PCO 1.2 Individuals with a family history of pancreatic cancer affecting two first-degree relatives meet criteria for familial pancreatic cancer. Individuals whose family history

TABLE 1. Genes Associated With Increased Risk for Pancreatic Cancer

Gene	Syndrome	Pancreatic Cancer Risk (%)	Other Associated Cancers*
<i>APC</i>	Familial adenomatous polyposis	1-5	Colorectal, upper GI, thyroid, brain
<i>ATM</i>	Ataxia telangiectasia (biallelic)†	1-5	Breast, prostate, gastric
<i>BRCA2</i>	Hereditary breast ovarian cancer syndrome	5-10	Breast, ovary, prostate, melanoma
<i>BRCA1</i>	Hereditary breast ovarian cancer syndrome	2	Breast, ovary, prostate, melanoma
<i>CDKN2A</i>	Familial atypical multiple mole melanoma (FAMMM)	10-30	Melanoma
<i>MLH1, MSH2, MSH6, PMS2, EPCAM</i>	Lynch syndrome	5-10	Colorectal, uterine, upper GI, ovary, urinary tract, brain, sebaceous neoplasms
<i>PALB2</i>		5-10	Breast, prostate
<i>STK11</i>	Peutz Jeghers syndrome	10-30	Breast, colorectal, upper GI, lung, reproductive tract
<i>TP53</i>	Li Fraumeni syndrome	Not defined	Breast, brain, sarcoma, adrenocortical carcinoma

*Most commonly associated cancers.

†Biallelic *ATM* mutation carriers have ataxia telangiectasia, but a single *ATM* mutation is associated with increased risk for pancreatic cancer.

meets criteria for familial pancreatic cancer, those with three or more diagnoses of pancreatic cancer in same side of the family, and individuals meeting criteria for other genetic syndromes (Table 1) associated with increased risk for pancreatic cancer have an increased risk for pancreatic cancer and are candidates for genetic testing (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

Qualifying statement. It is important to note that for 90% of families meeting criteria for familial pancreatic cancer, genetic testing does not detect a pathogenic mutation; therefore, there may be additional shared epigenetic, genetic, or environmental factors that contribute to pancreatic cancer risk.

PCO 1.3 Genetic risk evaluation should be conducted in conjunction with health care providers familiar with the diagnosis and management of hereditary cancer syndromes to determine the most appropriate testing strategy and discuss implications of the findings for family members. Germline genetic testing for patients with pancreatic cancer should be offered in the context of shared decision making⁴⁴⁻⁴⁷ (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

Research Question 2

Which individuals should undergo genetic testing for predisposition to pancreatic cancer?

PCO 2.1 All patients diagnosed with pancreatic adenocarcinoma should undergo assessment of risk for hereditary syndromes known to be associated with an increased risk for pancreatic adenocarcinoma (Table 1). Assessment of risk includes obtaining a personal cancer history and family history of cancers in first- and second-degree relatives. However, recent data demonstrate that many individuals who develop pancreatic cancer in the setting of

genetic predisposition lack clinical features or family cancer history typically associated with the corresponding hereditary syndrome. Therefore, germline genetic testing may be discussed with patients with personal history of pancreatic cancer, even if family history is unremarkable (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

Rationale. Recent studies have identified pathogenic germline variants in cancer predisposition genes in 4% to 20% of patients with pancreatic cancer.^{2,8,12,21,30,42,44-46,48-50} The prevalence of germline mutations is especially high among individuals of Ashkenazi Jewish ancestry, as a result of the high prevalence of *BRCA1* and *BRCA2* founder mutations in this population. Approximately half of patients with pancreatic cancer found to carry pathogenic genetic variants have no family history of pancreatic cancer and/or do not meet clinical criteria for the hereditary syndrome corresponding to their genetic diagnosis; therefore, patients should not be precluded from an opportunity to benefit from genetic testing just because their personal or family cancer history does not provide positive evidence of an inherited cancer syndrome. Given the potential that this information could be of benefit to the patient directly or to their family, the option of genetic testing should be discussed in the context of shared decision making. Germline genetic testing may not be indicated for patients with incurable pancreatic cancer who lack any living family members or for whom the results will not otherwise change management.

PCO 2.2 An individual with a cancer diagnosis is often the best candidate in whom to initiate genetic testing and has the highest likelihood of an informative test result; however, if a cancer-affected individual is not available, testing may be performed in a pancreatic cancer-unaffected individual following genetic risk assessment, with the understanding that a negative test result is considered clinically uninformative.

The following cancer-unaffected individuals should be offered genetic risk evaluation:

- Members of families with an identified pathogenic cancer susceptibility gene variant
- Pancreatic cancer–unaffected individuals from families that meet criteria for genetic evaluation for known hereditary syndromes that are linked to pancreatic cancer
- Pancreatic cancer–unaffected individuals from families that meet criteria for familial pancreatic cancer, as outlined in PCO 1.2

(Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

PCO 2.3 Genetic testing in a patient with pancreatic cancer may confirm the diagnosis of a hereditary cancer syndrome and inform management of at-risk family members. Given the possibility that certain germline variants could potentially be used to guide therapeutic decision making and the limited prognosis of many patients with pancreatic cancer, the Expert Panel recommends that consideration of germline testing for inherited cancer susceptibility should be performed early in the disease course for patients with pancreatic cancer (Type: informal consensus; relative balance of benefits and harms; Strength of statement: moderate).

PCO 2.4 Several genes have been linked to risk for pancreatic cancer (Table 1). Unless a genetic diagnosis has previously been confirmed in a family member, germline genetic testing should be performed using a multigene panel that includes the genes listed in Table 1. A finding of a pathogenic or likely pathogenic germline variant can confer increased risks of cancers beyond the pancreas for the probands and their families; finding a germline variant of uncertain significance is not considered to be causative of increased cancer susceptibility^{47,51} (Type: informal consensus; benefits outweigh harms; Strength of statement: strong).

Research Question 3

What surveillance strategies should be used for individuals with predisposition to pancreatic ductal adenocarcinoma to screen for pancreatic and other cancers?

PCO 3.1 Pancreatic cancer surveillance can be considered for individuals who are first-degree relatives of individuals with familial pancreatic cancer^{2,5,34,37,52} and/or individuals with a family history of pancreatic cancer who carry a pathogenic germline variant in genes associated with predisposition to pancreatic cancer (Table 1). The potential risks, benefits, uncertainties, and limitations of surveillance for pancreatic cancer should be discussed in detail with individuals who are being considered for pancreatic cancer surveillance prior to beginning such surveillance. When possible, pancreatic surveillance should be performed at centers with the appropriate expertise to manage individuals at increased risk for pancreatic cancer.

Surveillance may be performed with various modalities, including pancreas protocol magnetic resonance imaging/magnetic resonance cholangiopancreatography and/or endoscopic ultrasound. There are currently no approved biomarkers for screening and surveillance. CA 19-9 is not recommended as a screening test in the general population because of low specificity and sensitivity; its potential utility in pancreatic screening of high-risk individuals has not been established (Type of recommendation: informal consensus; relative balance of benefits and harms; Strength of recommendation: moderate).

Qualifying Statement. Although large studies confirming mortality benefit of pancreatic screening are lacking, emerging data suggest screening in individuals with high risk is associated with downstaging of incident cancers.⁵³

PCO 3.2 There is not yet consensus on pathologic targets for surveillance, but the Expert Panel agrees the ultimate goal should be detection and treatment of high-grade dysplasia to prevent invasive cancer (Type: informal consensus; relative balance of benefits and harms; Strength of statement: moderate).

Rationale. Although survival of patients with screen-detected cancers is improved relative to those who present with symptomatic cancers, many patients with screen-detected pancreatic cancers will die as a result of their disease.^{40,53} In individuals who carry genetic alterations and families with a high prevalence of pancreatic cancer, the prevalence of pancreatic abnormalities (mostly small pancreatic cystic neoplasms of low-malignant potential) is higher than in the general population.^{35,54} Early experience of pancreatic screening programs indicates that pancreatic cancers detected by screening are downstaged.^{40,53}

PCO 3.3 The potential risks of surveillance, including the risk of overtreatment and unnecessary resections, should be discussed with the patient. Given the challenges, patients should optimally be managed by an expert multidisciplinary team with experience in pancreatic cancer surveillance. Additional clinical studies are needed to determine the optimal approach for pancreatic surveillance (Type: informal consensus; relative balance of benefits and harms; Strength of statement: moderate).

PCO 3.4 There is currently a lack of consensus regarding which lesions discovered by pancreatic imaging require resection. Findings that generally warrant resection include lesions that are solid and/or that are associated with obstructive jaundice and/or dilation of the main pancreatic duct greater than 10 mm. The presence of worrisome features (cyst size greater than 3 cm, thickened/enhancing walls, mural nodule, dilated main pancreatic duct greater than 5 mm, abrupt change in duct caliber, or rapid growth) and the presence of three or more pancreatic cysts in the pancreas of high-risk individuals are associated with an increased risk of neoplastic progression.⁵³

PATIENT-CLINICIAN COMMUNICATION

Cancer-unaffected individuals with a relative diagnosed with pancreatic cancer face the uncertainty of knowing if they too are at risk for the disease. Communication between patients and clinicians is key to understanding the cancer-unaffected individual's concerns when making decisions about pursuing genetic testing. Cancer-unaffected individuals should fully understand that this risk depends on the extent of their family history and their own medical history and can learn more about their risk by consulting with a clinician with cancer genetics expertise. They should also be aware that such an evaluation may help them better understand their overall cancer risk, but strategies to prevent pancreatic cancer, such as early detection, have not been proven to prevent pancreatic cancer. For cancer-unaffected individuals to make informed decisions, clinicians should describe the potential impact (both medical and emotional aspects) of genetic testing on both the individual and the family.

Cancer-unaffected individuals should also understand that many of the germline mutations associated with an increased risk of pancreatic cancer are also associated with increased risk of other cancers that exceed the lifetime risk for pancreatic cancer. Providers should share that there are effective strategies for prevention and screening of many of these other cancers, such as breast and colorectal cancer, associated with highly penetrant hereditary cancer syndromes (eg, Lynch syndrome, hereditary breast and ovarian cancer) identified in some families affected with pancreatic cancer, so genetic testing is reasonable for cancer-unaffected individuals meeting specific criteria.

It is advisable to provide resources to help patients communicate better with their health care team. Patients should be offered decision-making tools and urged to write down questions in between and in advance of appointments. Patients can be referred to resources that will extend the support and information clinicians are able to provide. For pancreatic cancer, two such resources are the ASCO patient-facing website (www.cancer.net) and the Pancreatic Cancer Action Network (www.pancan.org).

LIMITATIONS OF THE RESEARCH AND FUTURE DIRECTIONS

Utility of Multigene Panels for Clinical Genetic Testing

The prevalence of pathogenic germline variants in genes associated with cancer predisposition is higher among individuals diagnosed with pancreatic cancer, and the diversity of genes implicated provides justification for the use of multigene panel tests (Table 1). The most common germline findings are pathogenic variants in *BRCA1* and *BRCA2*, which have been associated with an estimated 3% and 5% lifetime risk for pancreatic cancer, respectively. *CDKN2A* germline mutations are also identified; in addition to risk of cutaneous melanoma, *CDKN2A* germline variants have been associated with increased risk of pancreatic

cancer, with relative risks between 52 and 80.8.^{55,56} By comparison, germline variants in *STK11* associated with Peutz Jeghers syndrome, while much rarer, are associated with a much more substantial increase in pancreatic cancer risk (relative risk, 132) as well as increased risk for other cancers.⁴ While the rationale for germline sequencing of genes associated with high-penetrance cancer syndromes is well established, the clinical utility of sequencing of moderate-penetrance genes has been challenged. Germline pathogenic variants in moderate-penetrance genes (eg, *ATM*, *CHEK2*) are found in one in every 100 individuals in the general population, but the magnitude of cancer risk increase associated with these pathogenic variants remains to be determined.⁵⁰ Additional genetic or environmental factors may modify cancer risk in individuals who carry pathogenic germline variants. A pathogenic or likely pathogenic variant is considered to have potential implications for patients with cancer and their families based on currently available surveillance, prevention, or treatment implications for any associated cancer (Table 1) in either the patient or his or her close family members. For example, many pancreatic cancer susceptibility genes are also included as high- and moderate-penetrance cancer genes in National Comprehensive Cancer Network screening and surveillance guidelines for breast and colorectal cancer.^{57,58} Patients who undergo genetic testing using a multigene panel should be counseled regarding the increased likelihood of finding a variant of uncertain significance.

Biomarkers

Despite considerable effort evaluating proteins, circulating tumor DNA, microRNA, exosomes, immunologic markers, and so on, there are currently no proven biomarkers using noninvasively obtained biospecimens (eg, blood, urine, stool) for early detection of pancreatic cancer in asymptomatic individuals.⁵⁹⁻⁶² A number of factors contribute to this challenging problem. Pancreatic cancer has a low overall annual incidence rate (12.5 per 100,000) and a 1.6% average lifetime risk in the United States⁶³; for this reason, even the most diagnostically accurate blood tests may be only suitable for those at increased risk of developing the disease. For this reason, recent efforts have focused on developing blood tests that can detect a panel of cancers so as to increase the pretest probability of a positive test and more comprehensive early cancer detection.⁶⁴ In addition, because of the expense and effort of follow-up imaging or invasive diagnostic testing, and the potential for psychological harm due to false-positive tests, only tests with high diagnostic specificity (approximately 99%) are suitable as early detection tests.

One challenge to the evaluation of candidate biomarker tests is the lack of prediagnostic biospecimens obtained from asymptomatic individuals with early-stage cancers.⁵⁹ Serum CA19-9 has been used as a biomarker of pancreatic cancer progression in patients but has limitations, because

around 10% of individuals are negative for Lewis antigen a or b (a-b-), and do not have detectable CA19-9.^{65,66} This reduces its value to use alone as a screening biomarker. Furthermore, CA-19-9 may also be elevated in patients with nonmalignant diseases, including liver cirrhosis, chronic pancreatitis, cholangitis, and other GI cancers.⁶⁷

There is no evidence on the clinical utility or validity of the use of circulating tumor DNA for screening outside of clinical trials.⁶⁸ Initial studies in the setting of pancreatic cancer screening indicate that the percentage of patients with stage I cancer who have detectable circulating tumor DNA is low, and its detection is associated with a poorer outcome.^{69,70} More recently, multimarker panels have been evaluated to improve biomarker diagnostic performance. In principle, combining multiple markers with high diagnostic specificity could improve diagnostic sensitivity, but high circulating tumor marker levels will often reflect higher tumor burden rather than early detection. Rigorous testing and validation of potential biomarkers, used either alone or in combination, in high-risk individuals, is a need in the field and an area of active investigation.

Another area of active investigation is understanding the relationship of pancreatic cancer diagnosis with a prior diagnosis of recent or new-onset diabetes (NOD) in individuals older than 50 years. There is a six- to eight-fold higher risk of being diagnosed with pancreatic cancer within 3 years of first meeting glycemic criteria for NOD, with a 3-year incidence of being diagnosed with pancreatic cancer of 0.5% to 1%.^{71,72} Notably, with the diagnosis of pancreatic cancer in the setting of new-onset diabetes is often accompanied by recent weight loss.^{73,74} Additional research is needed in this area, as there are currently no published studies evaluating the utility of an NOD diagnosis in individuals with an inherited/familial predisposition to pancreatic cancer.

Thresholds for Surgical Resection

Solid lesions of the pancreas, in particular those that are concerning to represent a possible adenocarcinoma of the pancreas, and other features that are considered high risk (ie, concerning for pancreatic cancer), such as obstructive jaundice, mural nodules of diameter 5 mm or more, and marked dilatation of the main pancreatic duct (diameter greater than 10 mm), generally require surgical resection. Worrisome features (eg, cyst size greater than 3 cm, thickened/enhancing cyst walls, small mural nodules, dilated main pancreatic duct 5 to 9 mm, abrupt change in duct caliber, and/or rapid cyst growth [greater than 5-mm increase in diameter within 2 years])⁵³ often represent advanced premalignant lesions, although the neoplastic grade of these lesions cannot be reliably determined by imaging. There is no consensus about which of these lesions in which patients are of sufficient concern to warrant pancreatic resection. Published reports on the indications and yield of pancreatic resection in the high-risk setting are

based on limited numbers of patients.^{35,75-78} Discussions about indications for surgical resection should be undertaken by experienced clinicians in a multidisciplinary setting. There are also many patients with minor pancreatic imaging abnormalities; the degree and extent of these varies considerably, and they often have uncertain neoplastic significance and are monitored. Pancreatic resection is not currently indicated in high-risk individuals without an identifiable pancreatic lesion.

HEALTH DISPARITIES

Although ASCO clinical practice guidelines represent expert recommendations on the best practices in disease management to provide the highest level of cancer care, it is important to note that many patients have limited access to medical care. Racial and ethnic disparities in health care contribute significantly to this problem in the United States. Patients with cancer who are members of racial/ethnic minorities suffer disproportionately from comorbidities, experience more substantial obstacles to receiving care, are more likely to be uninsured, and are at greater risk of receiving care of poor quality than other Americans.⁷⁹⁻⁸² For patients with pancreatic cancer specifically, it has been known for more than a decade that African Americans have a higher incidence and a higher risk of presenting with advanced-stage disease and are less likely to undergo surgical treatment of resectable disease.^{83,84} These racial disparities do not appear to have improved in recent years.⁸⁵ There are also substantial regional variations in utilization of multimodality therapy in treatment of pancreatic cancer.⁸⁶ Awareness of these disparities in access to care should be considered in the context of this provisional clinical opinion, and health care providers should strive to deliver the highest level of cancer care to these vulnerable populations.

COST CONSIDERATIONS

Increasingly, individuals with cancer are required to pay a larger proportion of costs through deductibles and coinsurance.^{87,88} Higher patient out-of-pocket costs have been shown to be a barrier to initiating and adhering to recommended cancer screening.^{89,90}

Discussion of cost should be an important part of shared decision making.⁹¹ Formal cost-effectiveness strategies for germline genetic testing in pancreatic cancer and pancreatic imaging in individuals at increased risk for pancreatic cancer are not available. However, given the substantial costs of diagnosis and treatment of pancreatic cancer as well as the lethality of the disease, early diagnosis is likely cost-beneficial to society. Indeed, the potential benefits of genetic testing can be extrapolated from results of cost-effectiveness studies of *BRCA* genetic testing in ovarian cancer, for instance.⁹² Genetic testing costs have diminished considerably but still can present a barrier to access, especially if not covered by third-party payers. A

transparent discussion about potential out-of-pocket costs of genetic testing as well as surveillance imaging should be conducted with patients and families.

EXTERNAL REVIEW AND OPEN COMMENT

The draft recommendations were released to the public for open comment March 30 through April 13, 2018. Response categories of “Agree as written,” “Agree with suggested modifications,” and “Disagree. See comments” were captured for every proposed recommendation in four questions. A total of five people responded to the survey questions with six written comments received. A total of 100% of the five respondents either agreed or agreed with slight modifications to the first section. A total of 100% of the five respondents either agreed or agreed with slight modifications to the second section. A total of 80% of the five respondents either agreed or agreed with slight modifications to the third section, and 20% disagreed. A total of 80% of the five respondents either agreed or agreed with slight modifications to the fourth section, and 20% disagreed. Expert Panel members reviewed comments from all sources and determined to revise the recommendations with minor language changes. All changes were incorporated prior to ASCO Clinical Practice Guidelines Committee review and approval.

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ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

ADDITIONAL RESOURCES

More information, including a Data Supplement with additional evidence tables, a Methodology Supplement with information about evidence quality and strength of recommendations, slide sets, and clinical tools and resources is available at www.asco.org/gastrointestinal-cancer-guidelines. Patient information is available at www.cancer.net.

Related ASCO Guidelines

- Locally Advanced Pancreatic Cancer⁹³ (<http://ascopubs.org/doi/10.1200/JCO.2016.67.5561>)
- Metastatic Pancreatic Cancer⁹⁴ (<http://ascopubs.org/doi/10.1200/JCO.2018.78.9636>)
- Potentially Curable Pancreatic Cancer⁹⁵ (<http://ascopubs.org/doi/10.1200/JCO.2017.72.4948>)

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST AND DATA AVAILABILITY STATEMENT

Disclosures provided by the authors and data availability statement (if applicable) are available with this article at DOI <https://doi.org/10.1200/JCO.18.01489>.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Evaluating Susceptibility to Pancreatic Cancer: ASCO Provisional Clinical Opinion**

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APPENDIX

TABLE A1. Hereditary Pancreatic Provisional Clinical Opinion Expert Panel Membership

Name and Designation	Affiliation/Institution	Role/Area of Expertise
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Elena M. Stoffel, Co-chair	University of Michigan, Ann Arbor, MI	Gastroenterology
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Gloria M. Petersen	Mayo Clinic, Rochester, MN	Epidemiology/genetics
Diane M. Simeone	New York University Langone Health, New York, NY	Surgical oncology
Marcia Canto	Johns Hopkins University, Baltimore, MD	Gastroenterology
Matthew Yurgelun	Dana-Farber Cancer Institute, Boston, MA	Medical oncology
Michael Goggins	Johns Hopkins University, Baltimore, MD	Gastroenterology
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Abbreviation: PGIN, Practice Guidelines Implementation Network.